

Statistical Machine Learning, EndSem, Total marks # 30

3rd May 2026

Instructions

1. Each MCQ carries 2.5 marks
2. MCQs - for each correct tick, you will get 0.5 marks. For each incorrect tick, you will lose 1 mark.
3. If you tick all correct answers, and, do not tick any incorrect answer, you will get full marks

Q1. [CO3] Which of the following statements about Random Forest are correct?

- (A) It trains multiple decision trees using bootstrap samples of the training data.
- (B) It usually combines predictions from different trees using majority voting for classification.
- (C) It always trains only one decision tree.
- (D) It can reduce variance compared to a single decision tree.
- (E) It selects a random subset of features while splitting nodes.

Q2. [CO2] Which of the following statements about Principal Component Analysis are correct?

- (A) PCA finds directions along which the projected data has maximum variance.
- (B) The first principal component corresponds to the direction of maximum variance in the data.
- (C) PCA always improves classification accuracy.
- (D) Principal components are mutually orthogonal.
- (E) PCA tries to maximize between-class separation.

Q3. [CO2] Which of the following statements about Fisher Discriminant Analysis are correct?

- (A) FDA uses class labels to find a projection direction.
- (B) FDA ignores class labels and only maximizes total variance.
- (C) In the two-class case, FDA projects data onto a single discriminant direction.
- (D) FDA and PCA always produce the same projection direction.
- (E) FDA tries to make samples from the same class compact after projection.

Q4. [CO2] Which of the following statements about Discriminant Analysis are correct?

- (A) Quadratic Discriminant Analysis can produce nonlinear decision boundaries.

- (B) Discriminant Analysis does not use class labels during training.
- (C) LDA commonly assumes that different classes share the same covariance matrix.
- (D) QDA allows different classes to have different covariance matrices.

Q5. [CO1] Consider a binary classification dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, where $\mathbf{x}_i \in \mathbb{R}^d, y_i \in \{-1, +1\}$. We want to learn two linear hyperplanes. Assume that the normal vectors of both hyperplanes are always unit vectors. Write an objective function that satisfies the following:

1. signed distance $d_1(\mathbf{x}_i)$ is close to the true label y_i ,
2. signed $d_2(\mathbf{x}_i)$ is close to the true label y_i ,
3. the two hyperplanes are orthogonal to each other.
4. the constraint that normal vectors are unit vectors need not be taken care of in the objective function. We just assume that they are unit vector but do not do anything about it.

Write the objective function and update rule for all parameters of one of the hyperplanes. [4]

Q6. [CO3] Consider the following scalar regression dataset with three training points: $(x_1, y_1) = (1, 1), (x_2, y_2) = (2, 3), (x_3, y_3) = (3, 5)$. In gradient boosting, instead of using the usual squared-error loss or absolute-error loss, suppose we use the following loss function $L(y, F(x)) = \exp(|y - F(x)|)$. The initial prediction is chosen as the average of the target values.

- (a) Derive the negative gradients. [1]
- (b) Using the negative-gradients formula, identify the modified targets [1]
- (c) Find next tree $h_1(x)$. Fit a regression tree with exactly one split using mean squared error. Consider the two possible splits: $x \leq 1.5$ and $x \leq 2.5$. Decide which split gives $h_1(x)$. In case both splits are equally good, use the smallest value of x for the split. [1]
- (d) If the learning rate is $\rho=1$, write the updated gradient boosted model and find the value predicted by this updated model for $x_1 = 1$. [1]

Q7. [CO1] Consider a neural network with - two input nodes x_1 and x_2 , one hidden node, one output node, no bias terms, absolute-value activation/non-linearity at the hidden node, no activation at the output node. Let the weights from the input nodes to the hidden node be w_1, w_2 . Let the weight from the hidden node to the output node be w_3 . Let the loss be MSE. Find an update rule for w_1 . [4]

Q8. [CO1] Show that the MLE estimate of variance of a Gaussian pdf is biased. [4]

Q9. [CO2] In Principal Component Analysis, the first principal component direction is obtained by maximizing the variance of the projected data. Now suppose we modify the PCA objective. Instead of maximizing the sum of squared projections, we maximize the sum of absolute projections while maintaining the constraint that the projection vector $u \in \mathbb{R}^d$ is unit vector. Find u . [4]